

1. Prove that $A \subseteq C$ and $B \subseteq C$ if and only if $A \cup B \subseteq C$.
2. Prove or disprove that $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
3. Prove or disprove that for all sets A and B , we have $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$.
4. Show that if a, b, x, y are elements, then $\{\{a\}, \{a, b\}\} = \{\{x\}, \{x, y\}\}$ if and only if $a = x$ and $b = y$.
What methods of proof are most convenient to use for each direction?